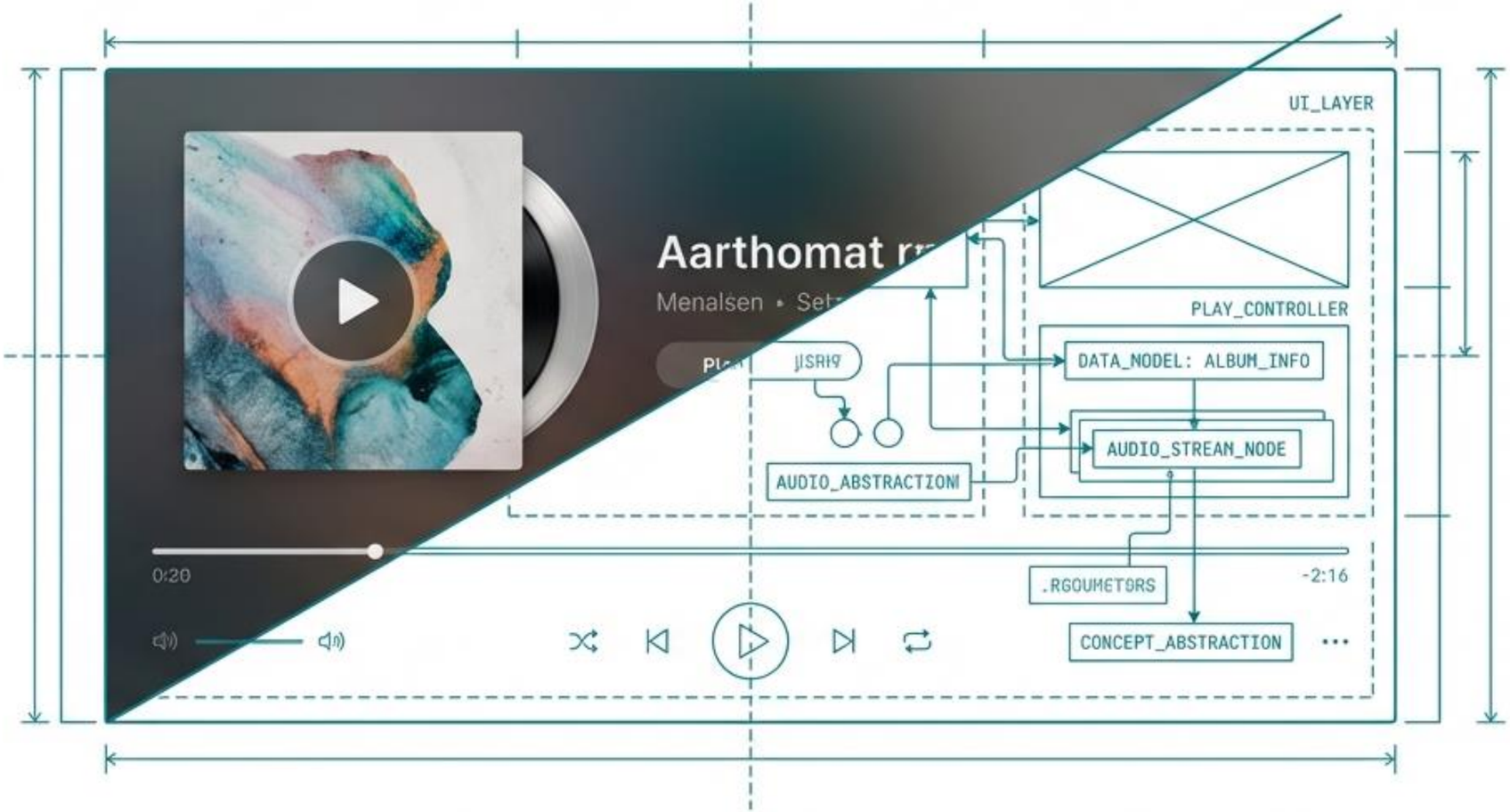
The Essence of Software

Why Concepts Matter for Great Design.



Based on the work of Daniel Jackson.

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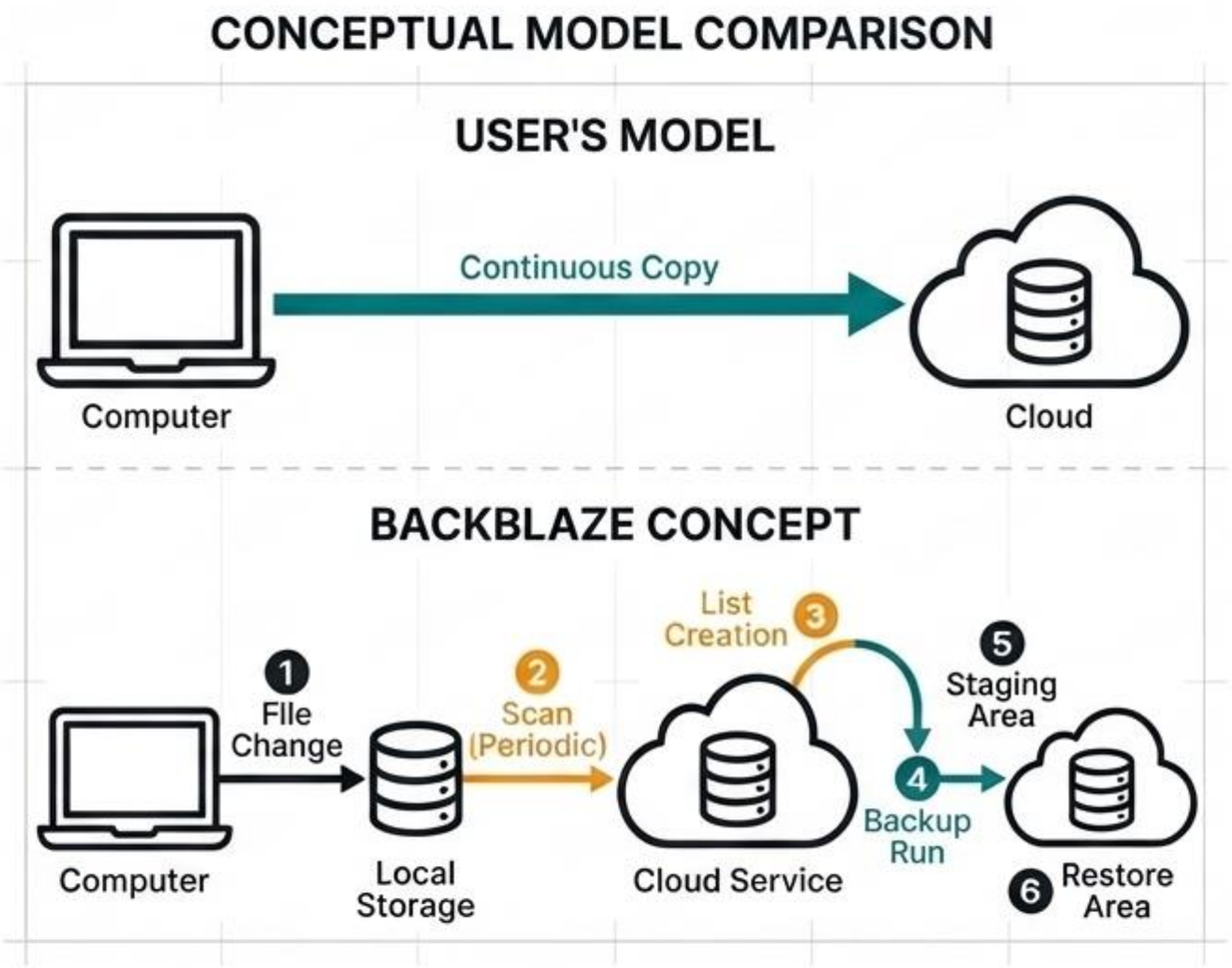
The Usability Paradox: Why "Simple" Apps Fail.

The Scenario: A user sees ‘Backed up’ status and deletes their local file to save space. They assume a ‘continuous copy’ model, where the cloud is a permanent archive, independent of the local state (as shown in ‘User’s Model’).

The Insight: The failure occurred because the system actually uses a ‘synchronization’ or ‘mirroring’ *concept* (as shown in ‘Backblaze Concept’). Deleting the local file triggered its deletion from the cloud, violating the user’s mental model.



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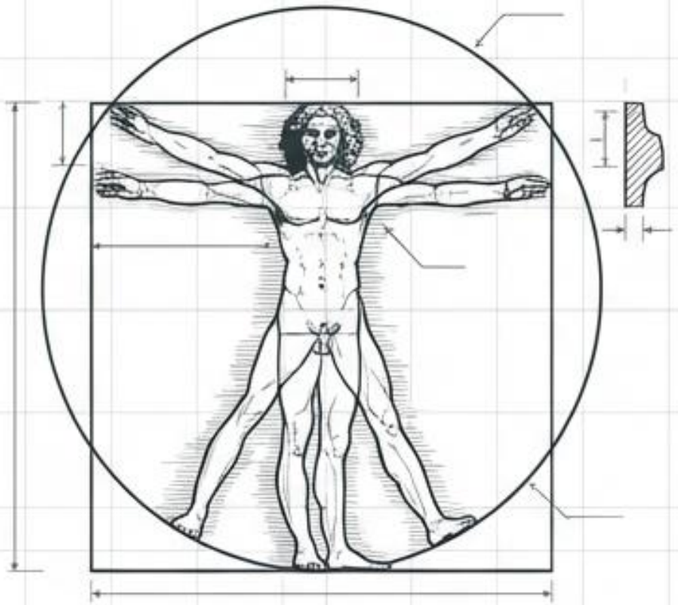
The Three Levels of Design.

Physical Level



Concrete. Concerned with color, size, layout, sound, and motor capabilities (Fitts's Law).

Example: Where is the menu bar placed?

Linguistic Level

Communication. Concerned with icons, labels, consistency, and culture.

Example: Do we call it a 'Folder' or a 'Directory'?

Conceptual Level

Abstract Essence. Concerned with semantics, actions, data models, and purposes.

Example: What is a 'Folder' and how does it behave when shared?

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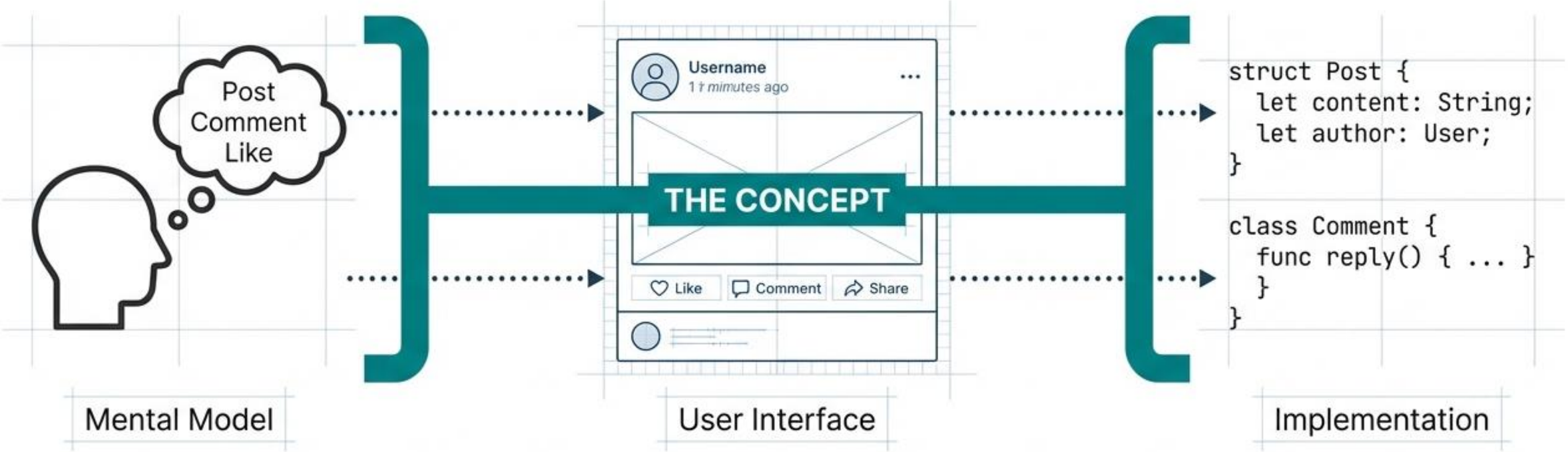
What is a "Concept"?

The Bridge Between Mind and Machine



Definition: A concept is a self-contained unit of functionality. It is the "molecule" of software—independent, reusable, and abstract.

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The Anatomy of a Concept.

HANDLE_ID

NAME

A handle for the concept. (e.g., *Trash*, *Style*, *Reservation*)

HANDLE_ID

PURPOSE

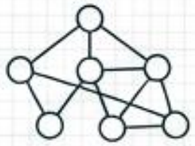
The specific human need the concept fulfills.

HUNAN_NEED_FULFILLMENT

STATE

The internal structure (sets, mappings) that tracks data.

INTERNAL_DATA_STRUCTURE



SETS_&_MAPPINGS

```
stroct ConceptState {
  itens: SetsItem>,
  settings: RapsKey, Value>
}
```

ACTIONS

Instantaneous behaviors that change the state. (e.g., *delete*, *restore*)

STATE_CHANGE_TRIGGER

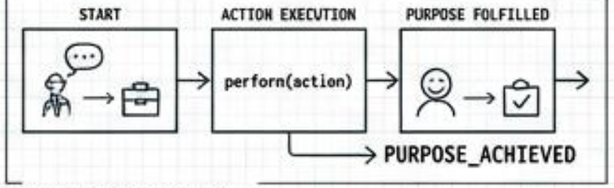
ACTION delete()

ACTION restore()

STATE_CHANGE_TRIGGER

OPERATIONAL_PRINCIPLE

The 'Script." A scenario demonstrating how actions fulfill the purpose.



SCENARIO_SEQUENCE

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Defining the Purpose: The “Why”

A purpose must be Cogent, Need-Focused, and Specific.

CONCEPT JetBrains Nono	DON'T JetBrains Nono	DO JetBrains Nono
Google Docs “Section” SECTION_BREAK	To break up ideas or set images apart from text.	To allow different margins and headers on different pages.
Browser “Bookmark” BOOKMARK_ID	To mark a page.	To make it easy to revisit a page later.
Apple Mail “VIP” VIP_SENDER		To keep track of messages from people important to you.

The Rule:
If you cannot articulate the purpose without circular logic, the design is likely flawed.

JetBrains Mono
DESIGN_RULE_001

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The Operational Principle: The “Script”

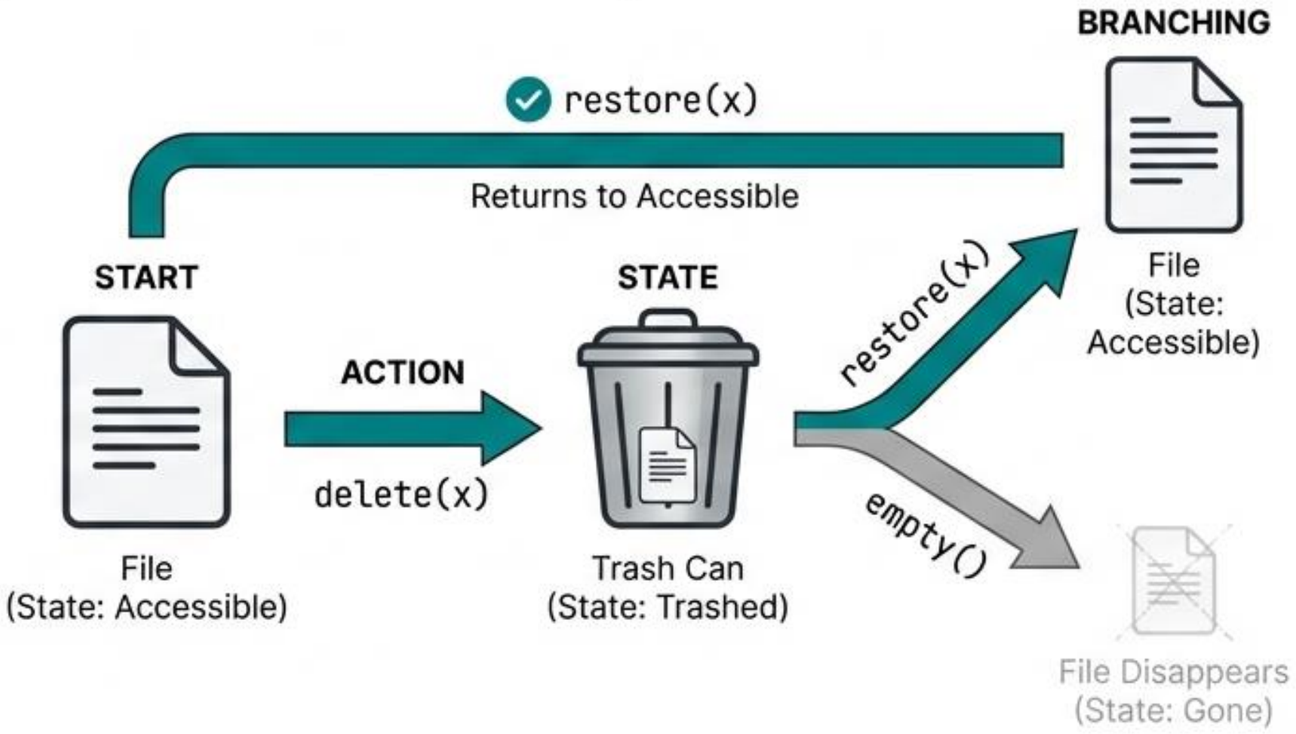
Case Study: The Trash Concept.

```
concept trash
  [Item]

purpose
  to allow undoing of deletions

state
  accessible, trashed: set Item

operational principle
  after delete(x), can restore(x)
  and then x in accessible.
```



The purpose of Trash is **NOT** deletion. It is **UN-deletion.**

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OPERATIONAL PRINCIPLE 'SCRIPT' FLOW
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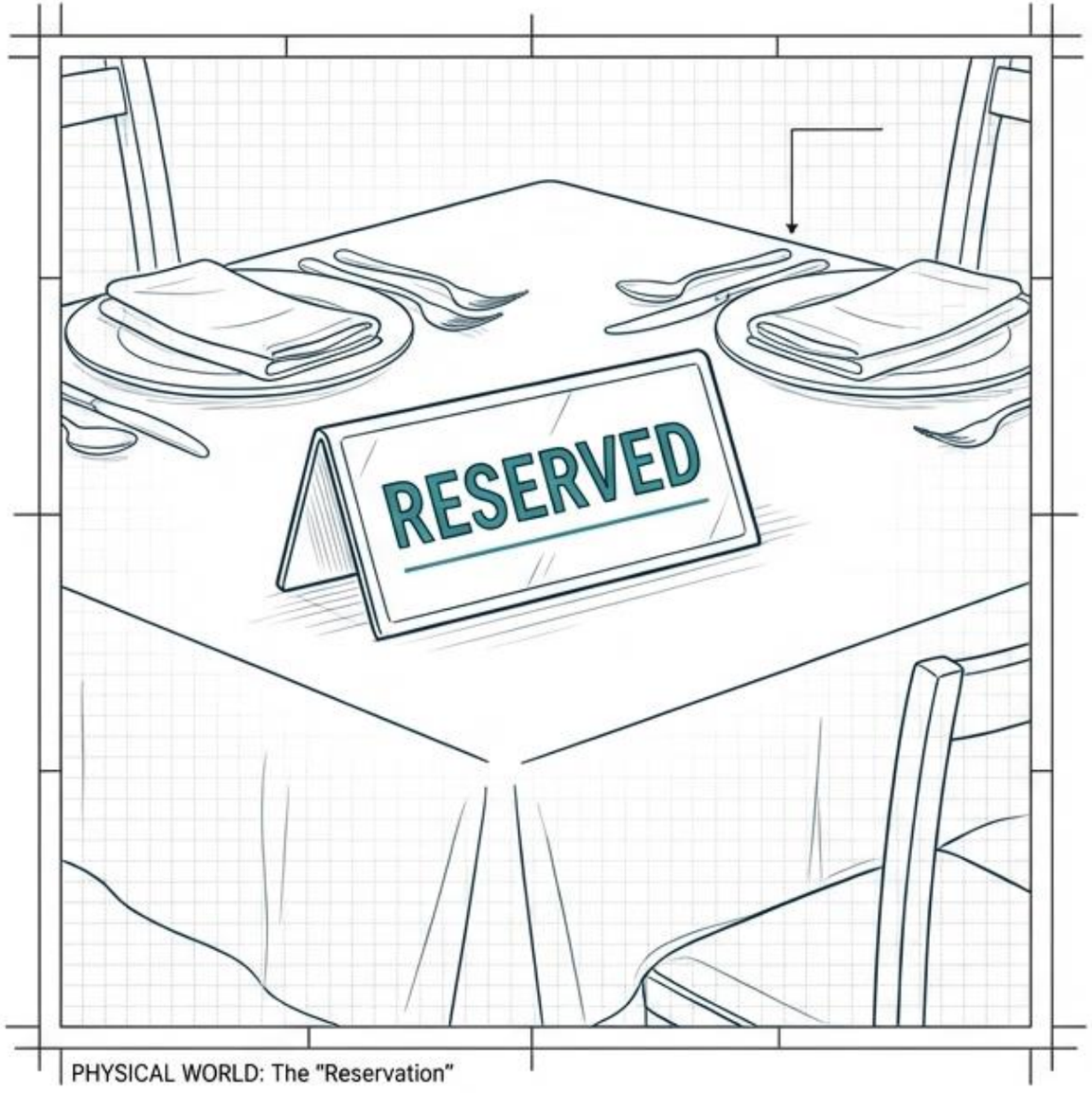
Concepts Predate Software: The “Reservation”



PHYSICAL WORLD: The "Reservation"

```
concept reservation [User, Resource]
 purpose manage efficient use of resources
 state   available: set Resource;
         reservations: User -> Resource
 actions reserve, cancel, use
```

The Hidden Complexities

- **Retraction:** Can the restaurant take the table back? (Physical vs. Logical State)
- **No-Shows:** What if you reserve but don’t use? (Resource Utilization Issue)
- **Conflict:** Managing overbooking (impossible in physics, allowed in software logic). (Logical Inconsistency Risk)

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CONCEPTS PREDATE SOFTWARE: RESERVATION

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Case Study: The “Dropbox” Confusion

Why did the shared file disappear?

User’s Mental Model: Tagging / Metadata

(Thinking the name is just a label on the file/folder itself.)

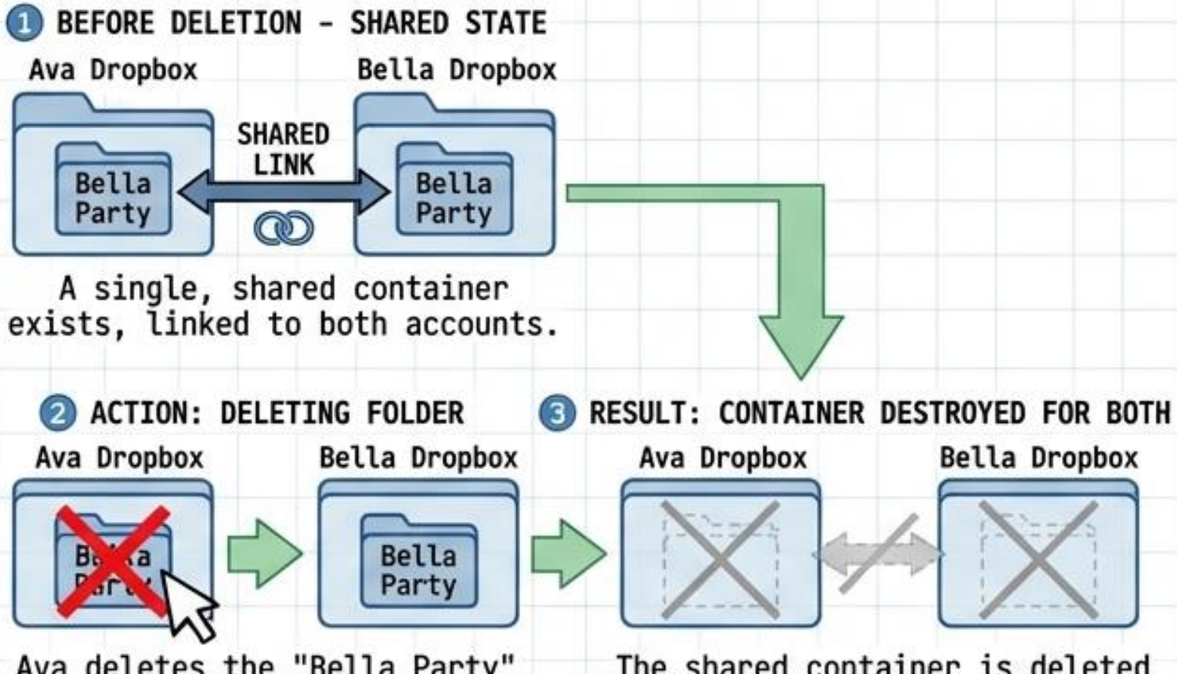
Ava thinks she has "tagged" a folder.

Ava removes the "Bella Party" label.

The folder object persists, just without the label.

Dropbox’s Technical Concept: Unix-style Folders

(The name is part of the directory structure, contained by the parent.)



A single, shared container exists, linked to both accounts.

Ava deletes the "Bella Party" folder.

The shared container is deleted from the system, removing it for Ava AND Bella.

Key Insight: Users expect a tag to be removed, but the action actually destroys the shared container. The model mismatch leads to data loss.

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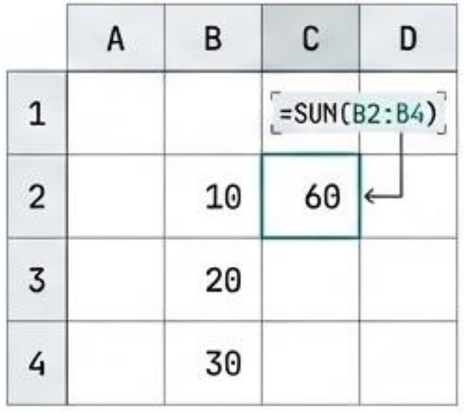
Concepts as Product Differentiators

Novel Concept + Critical Need = Market Dominance.

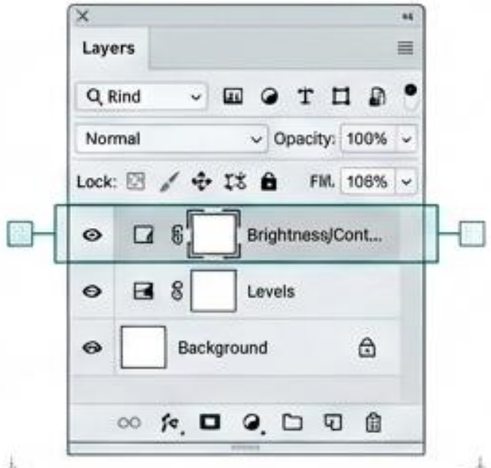
VisiCalc / Excel



The Formula
(Modeling Engine)

Allows real-time calculation and "what-if" analysis by linking cells dynamically. Changing one input instantiy updates all dependent formulas, enabling complex financial modeling.

Photoshop



The Layer
(Non-destructive editing)

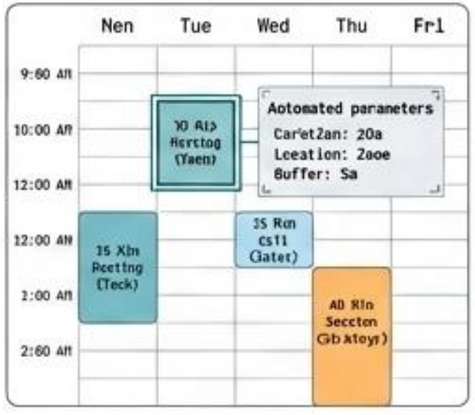
Enables independent adjustments and stacking of image elements without altering the original pixels. Edits are reversible and re-editable at any time, preserving quality.

World Wide Web

The URL
(Global persistent naming)

Provides a unique, universal address to locate and link resources across the internet. This simple identifier is the foundation of hypertext navigation and global information sharing.

Calendly



The Event Type
(Automated parameters)

Define fixed duration, location, and buffer times to automate scheduling. This standardizes the meeting format and eliminates back-and-forth coordination by offering pre-configured slots.

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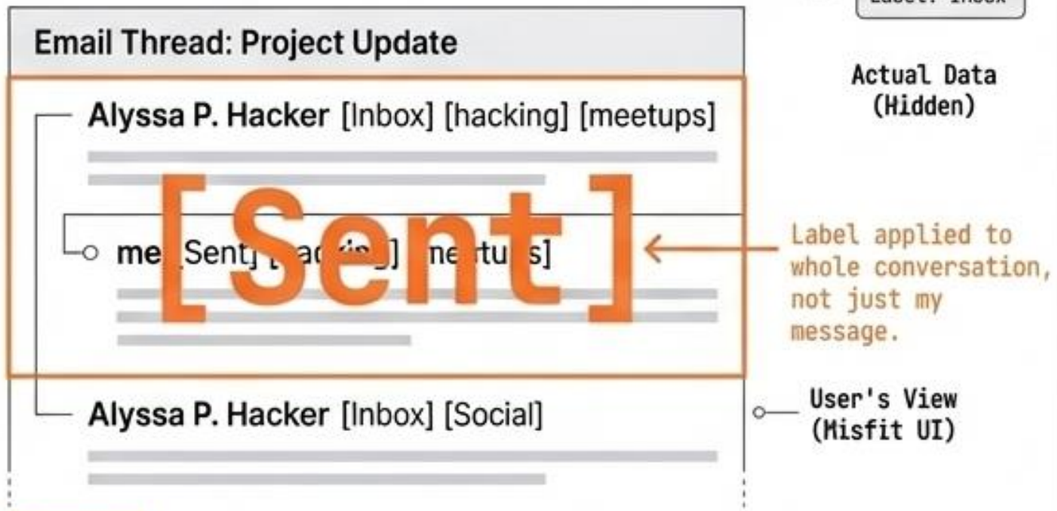
When Concepts Misfit

The cost of bad conceptual alignment.

Case 1: Gmail Labels

Concept A conversation is a thread of messages, each with its own labels (like 'Sent' or 'Inbox').

Misfit The UI treats 'Sent' as a state of the entire thread, not just the individual message. This hides who said what.

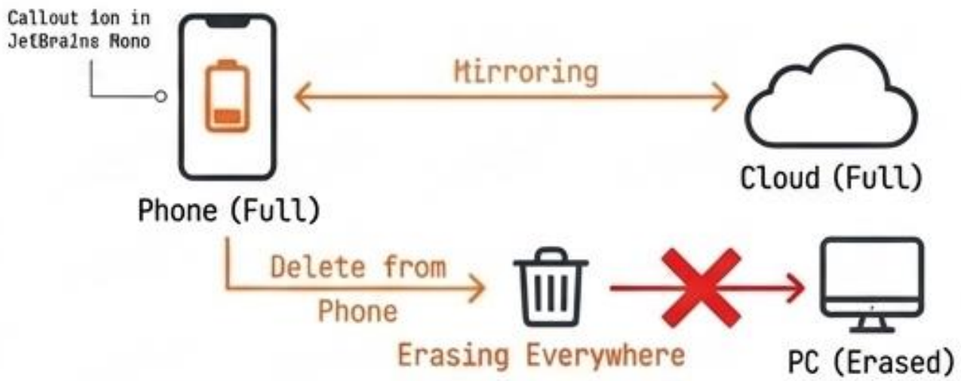


Result Users can't easily distinguish who said what in a conversation, leading to confusion.

Case 2: Apple Cloud Sync

Concept Mirroring (Keep exact same state on all devices).

Misfit The Catch-22. If one device is full, you can't free up space by deleting items, as it erases them everywhere.



Missing Concept Selective Sync.

Allow different states on different devices.

DRAWING TITLE: WHEN CONCEPTS MISFIT

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A Designer's Checklist

1. **Identify:**
   Can you name the core concepts of your app?

2. **Audit Purpose:**
   Is the purpose specific and user-focused? (Avoid "Manage X").

3. **Check the Script:**
   Write the Operational Principle. Does `Action A + Action B = Expected State`?

4. **Scrutinize Misfits:**
   Unix Folder in JetBrains Mono → Misfit → Sticky Label in JetBrains Mono
   Are you using a "Unix Folder" when the user expects a "Sticky Label"?

5. **Embrace Micromania:**
   → → Delete
   → Result for others?
   Obsess over edge cases. What happens when I delete a shared item?

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A DESIGNER'S CHECKLIST		
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The Essence of Software

To design software is to design a new world of concepts. Choose them wisely.

Software is not just code or pixels.
It is a system of Concepts.

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